Linear Algebra Project

Theory Concepts and Definitions

Students' Performance Analysis using Vectors, Matrices, Geometry and Eigen Decomposition

Item	Detail
Dataset	student_scores.csv (600 students × 6 subjects, synthetic)
Subjects	Maths, Physics, Chemistry, English, Computer Science, Statistics
Score range	0 to 100 per subject
Tools	Python (NumPy, pandas, Matplotlib)
Coverage	Part A: Vectors | Part B: Matrices | Part C: Geometry | Part D: Eigenvalues

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1. Introduction

A dataset of student scores is naturally a table of numbers. Linear algebra gives a precise language to store, transform and interpret such tables. Each student becomes a **vector**, the whole class becomes a **matrix**, and relationships between subjects are studied through the **covariance matrix** and its **eigenvectors**.

Definition – Scalar. A single real number, such as one score of 73 in Maths.

Definition – Vector. An ordered list of n numbers, written $v = (v_1, v_2, \ldots, v_n)$, which is a point (or an arrow from the origin) in n-dimensional space R^n.

Definition – Matrix. A rectangular array of numbers with m rows and n columns, called an m × n matrix. Our data matrix X is 600 × 6: rows are students and columns are subjects.

Dataset mapping

Mathematical object	Meaning in this project
Vector in R^6	Scores of one student, e.g. (73, 66, 75, 68, 71, 70)
Matrix X (600 × 6)	All students × all subjects
Axis / coordinate	One subject
Point in space	One student
Covariance matrix (6 × 6)	How subjects vary together

2. Part A: Vector and Matrix Fundamentals

2.1 Representing students as vectors

The first student has scores u = (73, 66, 75, 68, 71, 70) and the second has v = (59, 59, 61, 74, 52, 56) for the order Maths, Physics, Chemistry, English, Computer Science, Statistics. Both are vectors in R^6.

2.2 Norms

Definition – Norm. A function that assigns a non-negative length (size) to a vector. A norm is zero only for the zero vector, scales with scalars ($||c\,v|| = |c|\,||v||$) and satisfies the triangle inequality $||u + v|| \le ||u|| + ||v||$.

Definition – Norm-1 (Manhattan norm). The sum of the absolute values of the components.

$$||v||_1 = |v_1| + |v_2| + \ldots + |v_n|$$

Definition – Norm-2 (Euclidean norm). The straight-line length of the vector from the origin.

$$||v||_2 = \sqrt{(\, v_1^{\,2} + v_2^{\,2} + \ldots + v_n^{\,2}\,)}$$

Example: for u: $||u||_1 = 73+66+75+68+71+70 = 423$ and $||u||_2 = \sqrt{29875} \approx 172.84$. Interpretation: Norm-1 is the total marks of the student; Norm-2 measures overall strength and

gives more weight to large scores.

2.3 Dot product and angle

Definition – Dot product (inner product). For two vectors of equal length, the sum of products of corresponding components. It is a scalar. Geometrically it equals ||u|| ||v|| cos θ.

$$u \cdot v = u_1 v_1 + u_2 v_2 + \ldots + u_n v_n$$

$$\cos\theta = (u \cdot v) / (\|u\|_2 \|v\|_2), \quad \theta = \cos^{-1}(\cos\theta)$$

Example: u · v = 25420, giving θ ≈ **7.45°**. A small angle means the two students have a very similar performance profile (cosine similarity close to 1). If u · v = 0 the vectors are orthogonal (perpendicular).

2.4 Cross product

Definition – Cross product. Defined only for vectors in R^3. For a and b it gives a new vector perpendicular to both, whose length equals the area of the parallelogram spanned by a and b.

$$a \times b = (a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1)$$

Since the cross product exists only in 3D, three subjects are selected: Maths, Physics and Computer Science. For the two students, a = (73, 66, 71) and b = (59, 59, 52): a × b = (−757, 393, 413) with ||a × b|| ≈ 947.66. Properties: a × b = −(b × a); a × a = 0; (a × b) · a = 0 and (a × b) · b = 0 (verified in the code).

2.5 Vector projection

Definition – Projection of u onto v. The component of u that lies along the direction of v. The remaining part u − $\text{proj}_v(u)$ is orthogonal to v.

$$\text{proj}_v(u) = ((u \cdot v) / (v \cdot v)) v$$

Example: scalar factor = 25420 / 22 003 ≈ 1.155 applied to v gives proj ≈ (68.17, 68.17, 70.49, 85.51, 60.09, 64.71). Meaning: how much of student 1's score pattern can be explained by student 2's pattern. Projection is the foundation of least-squares fitting and PCA.

3. Part B: Matrix Operations

The data matrix X has 600 rows (students) and 6 columns (subjects). For square-matrix operations, two 6 × 6 blocks are taken: A = students 1–6 and B = students 7–12.

3.1 Addition and multiplication

Definition – Matrix addition. Defined for matrices of the same size; it adds corresponding entries: $(A + B)_{ij} = A_{ij} + B_{ij}$.

Definition – Matrix multiplication. For A (m × n) and B (n × p), the product AB is m × p with $(AB)_{ij} = \Sigma_k A_{ik} B_{kj}$ (row i of A dotted with column j of B). The number of columns of A must equal the number of rows of B. In general AB ≠ BA.

3.2 Transpose

Definition – Transpose. The matrix A^T obtained by swapping rows and columns: $(A^T)_{ij} = A_{ji}$. Rules: $(A^T)^T = A$, $(AB)^T = B^T A^T$.

3.3 Determinant

Definition – Determinant. A scalar attached to a square matrix. It tells the factor by which the matrix scales volume and whether the matrix is invertible (det ≠ 0) or singular (det = 0).

$$2 \times 2 \text{ case: } \det[[a, b],[c, d]] = ad - bc$$

For the 6 × 6 block A, **det(A) = −8 391 734**, which is non-zero, so A is invertible and has full rank 6.

3.4 Inverse

Definition – Inverse matrix. For a square matrix A, the matrix A^{-1} satisfying $A\,A^{-1} = A^{-1}\,A = I$ (identity matrix). It exists only if det(A) ≠ 0.

$$A^{-1} = \text{adj}(A) / \det(A)$$

The code confirms numerically that $A \times A^{-1} = I$. The full 600 × 6 matrix X is not square, so it has no determinant or ordinary inverse. Two substitutes are used:

- **Gram matrix** $X^T X$ (6 × 6): square and invertible here (rank 6).
- **Moore–Penrose pseudo-inverse** X^+ (6 × 600), a generalised inverse used in least squares.

Definition – Rank. The number of linearly independent rows (or columns) of a matrix. A square n × n matrix is invertible exactly when its rank is n.

Operation	Condition	Result for this dataset
A + B	Same size	6 × 6 matrix of summed scores
A × B	Columns of A = rows of B	6 × 6 matrix
A^T	Always defined	Subjects become rows
det(A)	Square matrix	−8 391 734
A^{-1}	det(A) ≠ 0	Exists ($A \times A^{-1} = I$)
$X^T X$	Always defined	6 × 6, rank 6

4. Part C: Linear Transformations and Geometry

4.1 Line, plane and hyperplane

Definition – Line. A one-dimensional set of points in a space, described in 2D by ax + by = c. In our data: using two subjects, a line such as Physics = m · Maths + c.

Definition – Plane. A two-dimensional flat surface in 3D, described by ax + by + cz = d. In our data: three subjects (Maths, Physics, Computer Science).

Definition – Hyperplane. A flat subspace of dimension n − 1 inside an n-dimensional space, described by w · x = b where w is the normal vector. In our data: the full 6-subject space contains 5-dimensional hyperplanes that can separate students into groups (for example above or below an average-score threshold).

$$\textit{Hyperplane:} \quad w_1x_1 + w_2x_2 + \ldots + w_nx_n = b$$

Results from the dataset (least-squares fits):

- Line: Physics = 0.800 × Maths + 9.869
- Plane: Computer Science = 0.485 × Maths + 0.295 × Physics + 20.092
- Hyperplane (6D): Statistics expressed as a weighted combination of the other five subjects.

Definition – Least squares. A method that chooses the line, plane or hyperplane minimising the sum of squared vertical distances between the fitted surface and the data points. It is solved using the normal equations $X^TX\,w = X^Ty$.

4.2 How dimensionality increases

Every new subject adds one coordinate axis. Each student remains one point, but the cloud of points lives in a higher-dimensional space and the hyperplane that cuts it also grows in dimension (always n − 1).

Subjects used	Space	Cutting object	Dimension of object
1 (Maths)	1D line	Point	0
2 (+ Physics)	2D plane	Line	1
3 (+ Comp. Science)	3D space	Plane	2
4 (+ Chemistry)	4D	Hyperplane	3
5 (+ English)	5D	Hyperplane	4
6 (+ Statistics)	6D	Hyperplane	5

Beyond three dimensions we cannot draw the space, but the algebra (dot products, norms, matrices) works exactly the same. This is why linear algebra is powerful for data with many features.

4.3 Linear transformation

Definition – Linear transformation. A function $T: R^n \to R^m$ such that T(u + v) = T(u) + T(v) and T(c u) = c T(u). Every linear transformation can be written as multiplication by a matrix: T(x) = M x.

Common examples are rotation, scaling, reflection and projection. In the code, a scaling of (1.2, 0.8) combined with a 30° rotation is applied to a (Maths, Physics) point; its determinant 0.96 is the factor by which areas change. Principal component analysis is itself a rotation of the data to new axes.

Definition – Linear independence and span. Vectors are linearly independent if none can be written as a combination of the others. The span of a set of vectors is all linear combinations of them. Six independent subject axes span R^6.

5. Part D: Eigenvalues, Eigenvectors and Decomposition

5.1 Mean, variance and covariance

Definition – Mean. The average of a subject's scores: $\bar{x} = (1/n)\ \Sigma\ x_i$.

Definition – Variance. The average squared deviation from the mean; it measures how spread out the scores are.

$$Var(x) = (1/(n-1))\ \Sigma\ (\ x_i - \bar{x}\ )^2$$

Definition – Covariance. A measure of how two subjects vary together. Positive: high scores in one go with high scores in the other. Negative: opposite. Zero: no linear relationship.

$$Cov(x, y) = (1/(n-1))\ \Sigma\ (\ x_i - \bar{x}\ )(\ y_i - \bar{y}\ )$$

Definition – Covariance matrix. A symmetric k × k matrix C whose diagonal entries are variances and whose off-diagonal entries are covariances. With centred data X_c = X − mean, it equals $C = X_c^T\ X_c\ /\ (n - 1)$.

In this dataset the largest variance is English (100.58) and the strongest covariance is between Maths and Statistics (74.22), showing that quantitative subjects move together while English behaves more independently.

5.2 Eigenvalues and eigenvectors

Definition – Eigenvector and eigenvalue. For a square matrix C, a non-zero vector v is an eigenvector if multiplying by C only stretches or flips it, not changes its direction: C v = λ v. The scalar λ is the corresponding eigenvalue.

$$C\ v = \lambda\ v \quad \Rightarrow \quad det(\ C - \lambda\ I\ ) = 0 \quad (characteristic\ equation)$$

How to find them manually:

- Form C − λI and set det(C − λI) = 0. This gives a polynomial of degree n in λ; its roots are the eigenvalues.
- For each λ, solve (C − λI) v = 0 to obtain the eigenvector direction.
- Normalise each eigenvector to length 1.

Key properties for a covariance matrix (symmetric, positive semi-definite):

- All eigenvalues are real and non-negative.
- Eigenvectors of different eigenvalues are orthogonal to each other.
- Trace(C) = sum of eigenvalues (total variance) = 542.764 here.
- det(C) = product of eigenvalues = 9.64×10^9 here.

Definition – Diagonalisation (spectral decomposition). A symmetric matrix can be written $C = Q\ \Lambda\ Q^T$, where the columns of Q are orthonormal eigenvectors and Λ is the diagonal matrix of eigenvalues.

5.3 Principal component analysis (PCA) and variance

Definition – Principal component. An eigenvector of the covariance matrix. The first principal component (PC1) is the direction of maximum variance, PC2 the next direction orthogonal to PC1, and so on. The eigenvalue equals the variance along that direction.

$$\text{Variance explained by } PC_k = \lambda_k / (\lambda_1 + \lambda_2 + \ldots + \lambda_n) \times 100\%$$

Component	Eigenvalue	Variance explained	Cumulative
PC1	359.228	66.18%	66.18%
PC2	85.244	15.71%	81.89%
PC3	34.032	6.27%	88.16%
PC4	26.814	4.94%	93.10%
PC5	21.098	3.89%	96.99%
PC6	16.349	3.01%	100.00%

5.4 Interpretation

- **PC1 (66.18%)** has positive loadings on every subject: it measures overall academic ability. Students with a high PC1 score do well across all subjects.
- **PC2 (15.71%)** has a large English loading (0.90) opposite in sign to the quantitative subjects: it separates verbal strength from quantitative strength.
- **PC1 + PC2 = 81.89%** of the variance, so the 6-dimensional data can be reduced to 2 dimensions with little loss of information.
- Small eigenvalues mean little variation in that direction; these directions can be dropped (dimensionality reduction).

Definition – Scree plot. A bar chart of eigenvalues (or variance explained) in descending order, used to decide how many principal components to keep. The 'elbow' after PC2 here suggests two components are enough.

6. Summary of Results and Glossary

6.1 Key numerical results

Task	Result
Norm-1 / Norm-2 of student 1	423 / 172.84
Dot product, student 1 · student 2	25 420
Angle between students	7.45°
Cross product (Maths, Physics, CS)	(−757, 393, 413)
Determinant of 6 × 6 block A	−8 391 734 (invertible)
Best-fit line	Physics = 0.800 Maths + 9.869
Best-fit plane	CS = 0.485 Maths + 0.295 Physics + 20.092
Largest eigenvalue	359.228 (66.18% of variance)
Trace of covariance matrix	542.764 = sum of eigenvalues

6.2 Quick glossary

Term	Short meaning
Vector	Ordered list of numbers; one student's scores
Matrix	Rectangular table of numbers; all students × subjects
Norm	Length or size of a vector
Dot product	Sum of component-wise products; measures alignment
Orthogonal	At 90°, dot product equal to zero
Cross product	Vector perpendicular to two 3D vectors
Projection	Part of one vector lying along another
Transpose	Matrix with rows and columns swapped
Determinant	Scalar that tests invertibility
Inverse	Matrix that undoes A: $A A^{-1} = I$
Rank	Number of independent rows or columns
Hyperplane	(n − 1)-dimensional flat subspace in n dimensions
Covariance matrix	Variances and pairwise covariances of subjects
Eigenvalue λ	Stretch factor along an eigenvector; variance along a PC
Eigenvector	Direction unchanged by the matrix: $C v = \lambda v$
PCA	Rotation of the data onto eigenvectors of the covariance matrix

Note: the dataset is synthetically generated for learning purposes. All numerical results quoted here match the output of the project Python code.